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VST: A Python-based deep learning tool for segmenting electron microscopy samples

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Summary

Volume Segmentation Tool (VST) is a Python based deep learning tool designed specifically to segment three-dimensional VEM biological data without extensive requirements for cross disciplinary knowledge in deep learning. The tool is made accessible through a user-friendly interface with visualisations and a one-click installer.

Recognising the current rapid expansion of the VEM field, we have built VST with flexibility and instance segmentation in mind, hoping to ease and accelerate statistical analysis of large datasets in biological and medical research contexts. VST is composed of two main parts: the PyTorch (Paszke et al., 2019)-based deep learning core that performs semantic/instance segmentation on volumetric grey scale image datasets, and a user interface that operates on top of it, responsible for constructing CLI commands to the core components for tasking. The general pipeline of VST is shown in Figure 1. We had put in efforts to ensure VST could automatically handle issues associated with large dataset sizes, instance segmentation, anisotropic voxels and imbalanced classes.

The initial performance testing for VST with various subjects has been reported (Huang et al., 2025). Further testing within postgraduate projects at the University of Otago, New Zealand has further established scale, being able to segment the entire mitochondrial complement of tumoursphers (Jadav et al., 2023), and to segment poorly demarked cell remnants within wool fibres (unpublished).

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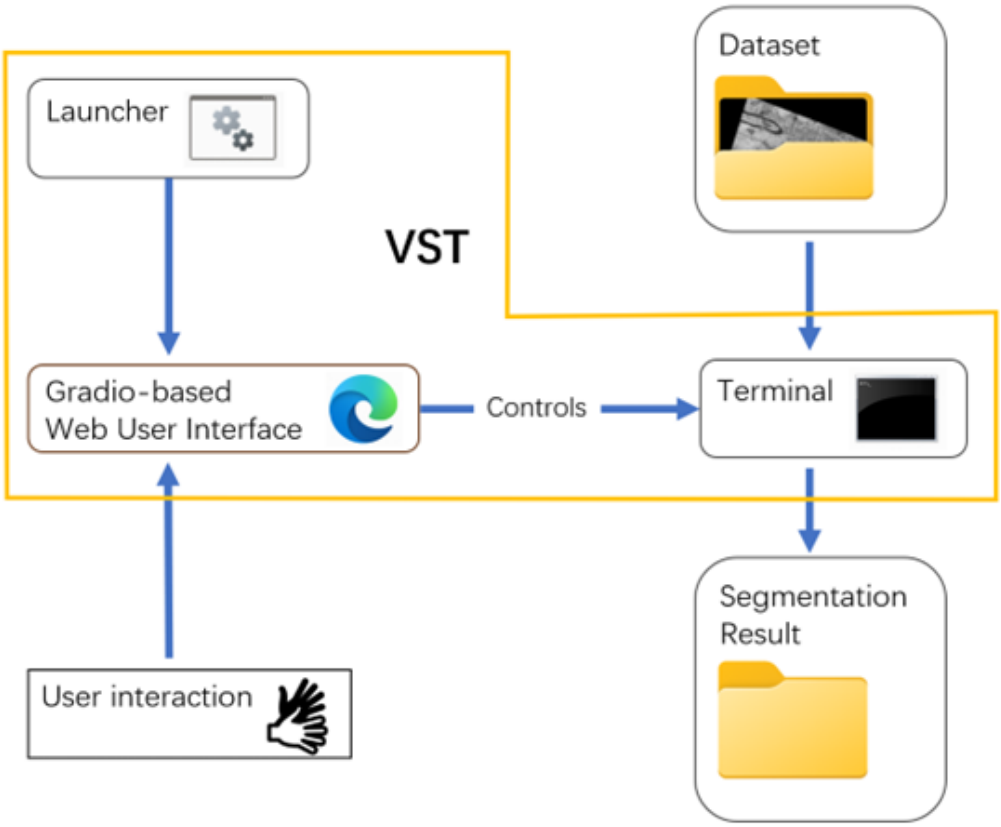

Figure 1: Schematic diagram for VST

Statement of need

Volume Electron Microscopy (VEM) enables the capture of 3D structure beyond planar samples, which is crucial for understanding biological mechanisms. With automation, improved resolution, and increased data storage capacity, VEM has led to an explosion of large three-dimensional datasets. Large datasets offer the opportunity to generate statistical data, but analysing them often requires assigning each voxel (3D pixel) to its corresponding structure, a process known as image segmentation. Manually segmenting hundreds or thousands of image slices is tedious and time-consuming. Computer-aided, especially Machine Learning (ML) based segmentation is now a routinely used method, with Trainable Weka Segmentation (Arganda-Carreras et al., 2017) and Ilastik (Berg et al., 2019) being two leading options. Emerging methods for EM image segmentation are often based on Deep Learning (DL) (Mekuč et al., 2020) because this approach has potential to outperform traditional ML in terms of accuracy and adaptivity (Erickson, 2019; Minaee et al., 2021).

Many earlier DL tools developed are highly specific to single sample types, like in connectomics (Kamnitsas et al., 2017; W. Li et al., 2017), MRI (Milletari et al., 2016) or X-ray tomography (A. Li et al., 2022), they use a subject-optimised design at the cost of adaptability to non-target datasets. Dedicated DL segmentation tools for generalised VEM data are gradually becoming available but each have short-comings. One example, CDeep3M (Haberl et al., 2018), which uses cloud computing. Although easy to use, it was designed for anisotropic data (where the z-resolution is much lower than xy-resolution) which creates limitations when applied to isotropic data (Gallusser et al., 2022). Another example is DeepImageJ (Gómez-de-Mariscal et al., 2021), which runs on local hardware and integrates easily with the ImageJ suit (Schneider et al., 2012). However, it only supports pre-trained models and does not have the functionality

to train new ones. ZeroCostDL4Mic (Von Chamier et al., 2021) utilises premade notebooks running on Google Colab, but it requires user interaction during the entire segmentation process, which can take hours and thus is inconvenient. A more recent and advanced example is nnU-Net (Isensee et al., 2021), which auto-configurates itself based on dataset properties and has a good support for volumetric dataset, but it focuses exclusively on semantic segmentation and lacks a user friendly interface.

In short, there is a lack of tools that can handle a wide range of VEM data well for generating both semantic and instance segmentation, while at the same time been easy to use, scalable and can be run locally. Which is what motivated us to develop VST - an easy-to-use and adaptive DL tools specifically optimised for generalised VEM image segmentation.

The graphical user interface

VST's GUI is supported by the Gradio package (Abid et al., 2019) and hosted on the user's browser.

The GUI is divided into three sections: Main, Activations Visualisation and Extras.

The main section (Figure 2) contains settings regarding training and using segmenting networks. Two segmentation modes are supported: semantic segmentation, in which the foreground objects are separated from the background, and instance segmentation, in which individual foreground objects are separated from each other as well. User can either train a new network, load an existing network and use it for predictions on new data, or train one and use it immediately.

Upon training, it automatically opens a Tensorboard interface (Pang et al., 2020) to provides various real time visualisations for the training process.

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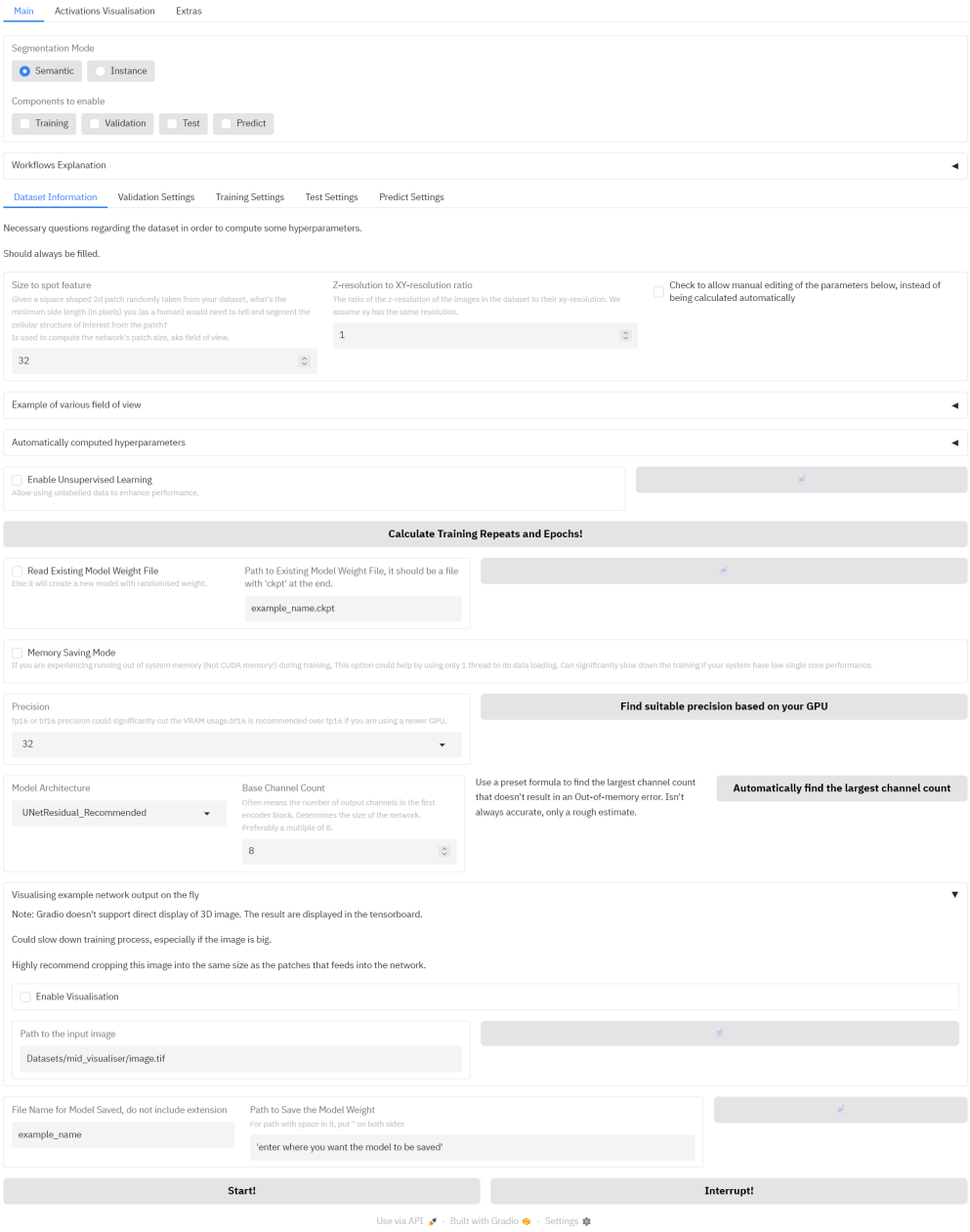

Figure 2: The main interface of VST

The activations visualisation section requires a trained network and an example image. Given that image, it plots the activation across each channel through all layers of the network.

The extra section contains two functionalities: exporting the TensorBoard log to an Excel table, calculating segmentation metrics between (potentially) generated labels and ground truth labels.

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